

Section 4 — Method: EvidenceFlow (中英对照)

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§4.1 Overview: Evidence Flow over a Query-Local Graph

总览：在查询局部图上的证据流

PARAGRAPH 1 — INPUT/OUTPUT + FORMATION PRINCIPLE

We treat multi-hop retrieval as the construction of a document-level evidence chain, rather than the independent ranking of documents. Given a question q , a corpus $\mathcal{D}$, and a pre-extracted fact substrate $\mathcal{F}$ derived from $\mathcal{D}$ during offline indexing, EvidenceFlow returns an evidence set $EK \subseteq \mathcal{D}$ of size K for the reader. The formation principle of EK is not that its documents are individually highly ranked against q , but that the set is constructed to prioritize connectivity through transitions that are witnessed by facts in $\mathcal{F}$.

我们把多跳检索视作构建一条文档级证据链的过程，而不是对文档独立打分排序。给定问题 q 、语料库 $\mathcal{D}$ ，以及离线索引阶段从 $\mathcal{D}$ 抽取出来的事实底层 $\mathcal{F}$ ，EvidenceFlow 为 reader 返回一个大小为 K 的证据集 $EK \subseteq \mathcal{D}$ 。 EK 的形成原则不在于其中的文档单独对 q 打分为多高，而在于该集合被构造为优先考虑通过 $\mathcal{F}$ 中事实背书的 **transition** 所建立连接性。

PARAGRAPH 2 — CORE MECHANISM (QUERY-LOCAL GRAPH + FACT-WITNESS ADMISSION)

The method operates on a *query-local document graph* $G_q=(V_q,A_q)$, instantiated per question rather than queried from a static pre-built knowledge graph. Vertices $V_q \subseteq \mathcal{D}$ are documents activated by the question, and an arc $(d_i,d_j) \in A_q$ is admitted only when facts in $\mathcal{F}$ provide a witness for the document-to-document movement implied by q . This admission rule is the central mechanism of EvidenceFlow: chain connectivity is a property of the retrieval graph itself, not a relation that a downstream module is asked to reconstruct over a flat passage pool.

方法在一张 **查询局部文档图** $G_q=(V_q,A_q)$ 上运行——这张图是为每个问题现拉的，而不是从一张静态预建的知识图谱里查出来的。顶点 $V_q \subseteq \mathcal{D}$ 是被该问题激活的文档；一条弧 $(d_i,d_j) \in A_q$ 只有当 $\mathcal{F}$ 中的事实能为该问题暗含的文档间移动提供 **witness** 时才被准入。这条准入规则就是 EvidenceFlow 的核心机制：链的连接性是检索图本身的属性，不是某个下游模块被要求在无类型 passage pool 上重新构建的关系。

PARAGRAPH 3 — FOUR CONSEQUENCES (FORWARD POINTERS)

Four consequences follow from this view. First , the fact substrate is built once at indexing time and reused across questions (§4.2); at query time, no language model is invoked to perform OpenIE extraction over or to generate evidence chains on retrieved text . Second , the activation of Gq is anchored by entities mentioned in the question, and the entry signal from dense or lexical retrieval only seeds this activation rather than dictating the final evidence (§4.3). Third , the movement of evidence through Gq is gated by fact witnesses, which makes each chain auditable back to source facts and their provenance in $\mathcal{D}$ (§4.4). Fourth , the reader receives EK under a chain-integrity boundary that prioritizes witnessed connectivity and admits residual documents when they provide evidence for still-open obligations induced by q (§4.5).	由此引出四个后果。第一，事实底层在索引阶段一次性建好，跨问题复用 (§4.2)； query 时不调用任何语言模型对 做 OpenIE 抽取，也不调用语言模型在检索到的文本上生成证据链。第二，Gq 的激活由问题里出现的实体来锚定，dense 或词项检索给出的入口信号只作为该激活的种子，而不决定最终证据 (§4.3)。第三，证据在 Gq 上的移动由 fact witness 把关，这使得每一条链都可以回溯到 中的源事实及其 provenance (§4.4)。第四，reader 在一个 链完整性边界 下收到 EK：这条边界优先考虑已被见证的连接性，当 residual 文档为 q 引出的尚未闭合义务提供证据时才被准入 (§4.5)。
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PARAGRAPH 4 — END-TO-END DECLARATION

Throughout, EvidenceFlow does not introduce a separate selection stage operating on a materialized candidate pool; the same query-local graph and the same fact-witness representation, together with the query-induced obligations that the boundary enforces, drive both how evidence is activated and how it is delivered to the reader.	贯穿整个流程，EvidenceFlow 不引入一个在已物化候选池上运行的独立选择阶段；同一张查询局部图、同一套 fact-witness 表征，加上边界处所执行的、由 query 引出的义务，既驱动证据如何被激活，也驱动证据如何被交付给 reader 。
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§4.2 OpenIE Fact Substrate

OpenIE 事实底层

PARAGRAPH 1 — SUBSTRATE DEFINITION + "WITNESS LAYER"

The fact substrate $\mathcal{F}$ stores triples of the form $f=(s,r,o)$ extracted from individual documents in $\mathcal{D}$, with each triple anchored back to its source document via a provenance index. $\mathcal{F}$ is constructed once during offline indexing and is reused unchanged across all questions. Its role in EvidenceFlow is not to replace documents as the unit of evidence: the reader still consumes natural-language passages from $\mathcal{D}$. The substrate is the witness layer for document-to-document movement, supplying the auditable links that justify why two documents may belong to the same evidence chain under a given question.	事实底层 $\mathcal{F}$ 存储形如 $f=(s,r,o)$ 的三元组，它们是从 $\mathcal{D}$ 中各篇文档抽取出来的，每个三元组通过 provenance 索引锚回其源文档。 $\mathcal{F}$ 在离线索引阶段一次性构造，跨所有问题原样复用。它在 EvidenceFlow 中的角色不是取代文档作为证据单元——reader 仍然消费来自 的自然语言文段。Substrate 是文档间移动的 witness 层，提供可追溯的连接，用以说明在给定问题下两篇文档为什么可以属于同一条证据链。
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PARAGRAPH 2 — EXTRACTION SETUP + CONTRIBUTION POSITIONING + PROTOCOL RED-LINE

Extraction follows the standard open information extraction setup used in prior graph-based retrieval systems (Angeli et al., 2015; Gutiérrez et al., 2025): a language model processes each document and emits its salient (s,r,o) assertions, which are then deduplicated and indexed alongside their source documents. We deliberately keep the extractor fixed: the contribution of EvidenceFlow is not a new fact extractor, but a different role for the facts it produces — as witnesses for document-level transitions rather than as the reasoning substrate or as prompt augmentation (§4.4). For the witness layer, this is the only corpus-level LLM extraction pass : after indexing, no query triggers fresh OpenIE extraction, fact repair, or chain generation over $\mathcal{D}$.	抽取遵循已有 graph-based 检索系统使用的标准 OpenIE 流程 (Angeli 等, 2015 ; Gutiérrez 等, 2025) : 一个语言模型逐文档处理并发出其显著的 (s,r,o) 断言, 这些断言随后被去重, 并连同其源文档一起索引。我们刻意保持抽取器不变: EvidenceFlow 的贡献不在于一个新的事实抽取器, 而在于其输出所担任的不同角色——作为文档级 transition 的 witness, 而非作为推理 substrate 或 prompt 增强 (§4.4)。 就 witness 层而言, 这是唯一的语料级 LLM 抽取 pass : 一旦索引完成, 任何 query 都不会触发对 $\mathcal{D}$ 的新一轮 OpenIE 抽取、事实修复或链生成。
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PARAGRAPH 3 — BOUNDARY AGAINST TOG / RELINK / NEOCORRAG

This separation of roles distinguishes EvidenceFlow from two adjacent lines of work. Knowledge-graph reasoning approaches (Sun et al., 2024; Huang et al., 2026) treat the extracted relations as the reasoning substrate itself — either by traversing a pre-built KG under LLM guidance (Sun et al., 2024) or by reconstructing missing facts at query time to repair broken paths (Huang et al., 2026); EvidenceFlow instead keeps $\mathcal{F}$ fixed and uses it only as a witness layer, so that retrieval does not depend on query-time fact generation or path repair. Query-time chain-mining approaches (Peng et al., 2026) keep documents as the reader unit but invoke a language model on the retrieved text to mine entity-relation chains as prompt augmentation; EvidenceFlow avoids this query-time mining step, and chain structure instead arises from the pre-extracted witnesses that gate document transitions in Gq (§4.4).	这种角色分工把 EvidenceFlow 跟两条邻近的工作线区分开来。知识图谱推理路线 (ToG, Sun et al. 2024 ; Relink, Huang et al. 2026) 把抽取出来的关系当作推理 substrate 本身 ——要么在预建 KG 上由 LLM 引导进行遍历 (ToG), 要么在 query 时重建缺失事实以修复断裂路径 (Relink) ; EvidenceFlow 则让 $\mathcal{F}$ 保持固定, 只作为 witness 层使用, 使得检索不依赖于 query 时的事实生成或路径修复。 Query 时链挖掘路线 (NeocorRAG, Peng et al. 2026) 保留文档作为 reader 单位, 但在检索到的文本上调用语言模型挖掘 entity-relation 链作为 prompt 增强 ; EvidenceFlow 避免这一 query 时挖掘步骤 , 链结构反而来自预抽取的 witness——这些 witness 在 §4.4 中把关 Gq 上的文档 transition。
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§4.3 Question-Anchored Document Graph

问题锚定的文档图

PARAGRAPH 1 — TWO COMPLEMENTARY SIGNALS

Given a question q , EvidenceFlow instantiates the query-local document graph $G_q=(V_q,A_q)$ by combining two complementary signals: a *question anchor* $A(q)\subseteq\mathcal{E}$ extracted from q , where $\mathcal{E}$ is the entity vocabulary induced by the fact substrate $\mathcal{F}$, and an *entry signal* $\text{Entry}(q)\subseteq\mathcal{D}$ obtained from a standard dense or lexical retriever. The anchor identifies which entities q is asking about; the entry signal identifies which documents are the most relevant starting points in $\mathcal{D}$. Neither signal alone determines the evidence delivered to the reader: V_q is initialized from these two signals and then formed where they can be connected through facts in $\mathcal{F}$.

给定问题 q ，EvidenceFlow 通过组合两类互补信号来实例化查询局部文档图 $G_q=(V_q,A_q)$ ：从 q 抽取出来的 **问题锚点** $A(q)\subseteq\mathcal{E}$ （其中 $\mathcal{E}$ 是由事实底层 $\mathcal{F}$ 诱导出的实体词表），以及由标准 dense 或词项检索器给出的 **入口信号** $\text{Entry}(q)\subseteq\mathcal{D}$ 。锚点指出 q 在问哪些实体；入口信号指出 $\mathcal{D}$ 里最相关的起点文档。任何一个信号单独都不决定交付给 **reader** 的证据： V_q 由这两类信号初始化，然后在它们能够通过 $\mathcal{F}$ 中事实相互连接的地方成形。

PARAGRAPH 2 — CONCRETE ACTIVATION

Concretely, V_q is initialized with the union of (i) documents in $\text{Entry}(q)$ and (ii) **documents that are the provenance of at least one fact in $\mathcal{F}$ whose arguments include an anchor entity in $A(q)$** . From this initial vertex set, V_q is expanded through a bounded number of hops along facts whose entities are connected to those already activated, with the size of V_q bounded to keep the activated region compact relative to $\mathcal{D}$, while the reader budget K ultimately constrains EK . This activation rule keeps $|V_q|$ small enough to make downstream computation tractable, while still allowing documents that are reachable only through multi-hop entity chains to enter V_q even when they have low surface similarity to q .

具体地， V_q 初始化为以下两类的并集：(i) $\text{Entry}(q)$ 中的文档；(ii) $\mathcal{F}$ 中至少存在一个事实，其 **argument** 包含 $A(q)$ 里某个锚点实体，而这些文档是该事实的 **provenance**。从这个初始顶点集出发， V_q 通过有界跳数沿事实扩展（只沿那些其实体跟已激活实体相连的事实）， V_q 的规模被约束以保持激活区域相对于 $\mathcal{D}$ 紧凑，而 **reader budget K** 最终约束 EK 。这一激活规则使 $|V_q|$ 小到足以让下游计算可行，同时仍然允许那些只能通过多跳实体链才能到达的文档进入 **V_q** ，即便它们对 q 的表面相似度低。

PARAGRAPH 3 — TWO PROPERTIES (ENTRY-SEED + QUERY-CONDITIONAL ANCHOR)

Two properties of this construction matter for the rest of the method. **First**, the entry signal seeds activation but does not by itself determine EK. Documents in Entry(q) enter Vq for consideration, but their position in the entry ranking is not the sole criterion that decides which of them reach the reader; what matters downstream is whether they participate in fact-witnessed transitions (§4.4). **Second**, the anchor extraction is the mechanism by which q exerts query-conditional control over Gq. Specifically, **anchors are extracted from entity mentions and relation-bearing expressions in q**, using the same controlled language model used throughout our protocol. **This extraction is applied to the question, not to the retrieved corpus, and does not generate evidence chains**; it provides the question-side semantic handles that allow Gq to grow toward documents that q actually asks about, rather than only toward documents that happen to score well on dense similarity.

该构造的两个性质对方法后续重要。第一，入口信号为激活播种，但不单独决定 **EK**——Entry(q) 中的文档进入 Vq 备选，但它们在入口排名中的位置并不是决定哪些到达 **reader** 的唯一准则；下游真正起作用的，是这些文档是否参与了由事实背书的 transition (§4.4)。第二，锚点抽取是 **q 对 Gq 施加 query-conditional 控制的机制**——具体地，锚点从 **q** 中的实体提及和承载关系的表达中抽取，使用整个协议中统一使用的同一个受控语言模型。这一抽取作用在问题上，不作用在检索到的语料上，也不生成证据链；它提供问题侧的语义抓手，让 Gq 朝着 **q** 真正在问的文档生长，而不是只朝着那些恰好在 dense 相似度上得分高的文档生长。

§4.4 Fact-Witnessed Evidence Transitions

事实背书的证据 transition

LEAD PARAGRAPH — CORE MECHANISM POSITIONING

We now turn to the core mechanism of EvidenceFlow: how the arcs Aq of the query-local graph are admitted. The aim is to make chain connectivity a structural property of Gq, audited at the moment of admission, so that **the evidence chain reaching the reader is constructed within the retrieval graph itself rather than reassembled from text afterward**.

现在转入 EvidenceFlow 的核心机制：查询局部图的弧 **Aq** 是如何被准入的。目标是让 chain 的连接性成为 Gq 的一个结构性属性——在准入那一刻就被审计——从而使得到达 **reader** 的证据链是在检索图内部构造出来的，而不是事后从文本中重新拼装出来的。

§4.4.1 WITNESS SETS

Let $f=(h,r,t)\in\mathcal{F}$ denote a fact extracted from some source document $\text{src}(f)\in\mathcal{D}$, with head and tail entities $h,t\in\mathcal{E}$. For an ordered pair of documents (d_i,d_j) , the *witness set* under question q is

$$W_q(d_i,d_j)=\{w\in\mathcal{M}(\mathcal{F})|w \text{ links } d_i \text{ and } d_j \text{ under } q\},(1)$$

where $\mathcal{M}(\mathcal{F})$ ranges over witness objects induced by facts in $\mathcal{F}$: **a single fact f whose endpoints reach across d_i and d_j , or a pair of facts (f_i,f_j) with $\text{src}(f_i)=d_i$ and $\text{src}(f_j)=d_j$ that share an activated argument.** A witness w links d_i and d_j under q when one of its arguments is activated by q (either through the anchor set $A(q)$ or through entities already entered the activated region of V_q) and a complementary argument grounds the other document. An arc (d_i,d_j) is *eligible* for A_q when $W_q(d_i,d_j)$ is non-empty; **surface similarity between d_i and d_j or their individual scores against q may seed activation but are not by themselves sufficient to create an arc.**

令 $f=(h,r,t)\in\mathcal{F}$ 表示从某个源文档 $\text{src}(f)\in\mathcal{D}$ 中抽取出来的一个事实，head 和 tail 实体 $h,t\in\mathcal{E}$ 。对一对有序文档 (d_i,d_j) ，在问题 q 下的 *witness 集* 定义为：

$$W_q(d_i,d_j)=\{w\in\mathcal{M}(\mathcal{F})|w \text{ 在 } q \text{ 下连接 } d_i \text{ 与 } d_j\},(1)$$

其中 $\mathcal{M}(\mathcal{F})$ 取值于由 $\mathcal{F}$ 中事实诱导出来的 witness 对象：可以是一个单一事实 f ，其端点跨越 d_i 和 d_j ；也可以是一对事实 (f_i,f_j) ， $\text{src}(f_i)=d_i$ 、 $\text{src}(f_j)=d_j$ ，且共享一个被激活的 **argument**。当 witness w 的某个 argument 被 q 激活（通过锚点集 $A(q)$ ，或通过已经进入 V_q 激活区域的实体），并且一个互补的 argument 锚定另一篇文档时， w 在 q 下连接 d_i 与 d_j 。一条弧 (d_i,d_j) 被准入 A_q 当且仅当 $W_q(d_i,d_j)$ 非空； **d_i 与 d_j 之间的表面相似度、它们各自对 q 的打分，可以为激活播种，但单独不足以建立一条弧。**

§4.4.2 AUDITABILITY

Every arc in A_q therefore carries an explicit witness in $\mathcal{F}$: each chain that ultimately reaches the reader can be traced back, edge by edge, to the source facts and their provenance in $\mathcal{D}$. **This is what distinguishes EvidenceFlow's notion of connectivity from a similarity graph in which two documents are joined whenever they share topic vectors:** similarity may motivate co-occurrence, but it does not establish that the two documents support a multi-hop step that q asks about. Witness sets provide this support by construction.

A_q 中的每一条弧因此都携带 $\mathcal{F}$ 中一个显式的 witness：每一条最终到达 **reader** 的链都可以被一边一边地回溯到其中的源事实及其 **provenance**。这就是 **EvidenceFlow** 的连接性概念跟相似度图（只要两篇文档共享 **topic vector** 就连一条边）的区别：相似度可以解释共现，但它无法证明两篇文档支持一个 **q** 真正在问的多跳步骤。Witness 集通过构造提供这种支持。

§4.4.3 WITNESS MODES

Within Eq. (1), two complementary modes of linkage naturally arise. A *sentence-grounded witness* is a single fact whose head and tail are both anchored in a source sentence that already mentions an entity activated by q ; this captures cases where one document explicitly bridges to the other through a q -relevant entity within a bridging sentence. An *argument-grounded witness* is induced by fact arguments grounded in the two documents through a shared activated entity, capturing cases where the multi-hop step is realized through entity coreference across documents rather than a single bridging sentence. Both modes share the same admission condition (a non-empty Wq) and the same auditability property; they differ only in how the witness is realized within $\mathcal{F}$.

在式 (1) 内，两种互补的连接模式自然涌现。句子锚定的 *witness* 是 head 和 tail 都被锚定在同一个源句子里的事实，且该句子已经提到了被 q 激活的某个实体——这刻画了 d_i (或 d_j) 中单个句子通过 q -相关实体显式桥接到另一篇文档的情况。Argument 锚定的 *witness* 由其 argument 锚定在两篇文档中、通过一个共享激活实体连接的事实诱导而来，刻画了多跳步骤通过跨文档的实体共指实现的情况（而非单一桥接句）。两种模式共享同一个准入条件 (Wq 非空) 和同一个可审计性属性；它们只在 witness 如何在 $\mathcal{F}$ 中实现的方式上不同。

§4.4.4 PRE-EXTRACTED RATHER THAN GENERATED WITNESSES

A crucial property of Eq. (1) is that **witnesses range over $(\mathcal{F})$, which is induced by the pre-extracted substrate from §4.2, and not over text retrieved at query time.** At the point of arc admission, the query-conditional information used to test witnesses is grounded in $A(q)$ and the current state of Vq ; witnesses themselves are not generated, paraphrased, or re-extracted from $\mathcal{D}$ for q . **Chain structure thus arises as a property of the retrieval graph rather than as a relation reconstructed from retrieved text at query time.**

式 (1) 的一个关键性质是：**witness** 在 $(\mathcal{F})$ 上取值——即从 §4.2 预抽取的底层诱导而来——而不是在 **query** 时检索到的文本上取值。在弧准入的时刻，用于检验 witness 的 query-conditional 信息基于锚点集 $A(q)$ 和 Vq 的当前状态；witness 本身不会为 q 而被生成、改写或对 重新抽取。链结构因此作为检索图的属性涌现，而不是在 **query** 时对检索到的文本重构出来的关系。

§4.5 Reader-Boundary Chain Integrity

Reader 边界处的链完整性

LEAD PARAGRAPH — BOUNDARY AS GRAPH-FORMATION OUTCOME

The query-local graph Gq constructed in §4.3–§4.4 typically contains more documents than the reader budget K admits. **At the reader boundary, EvidenceFlow forms the evidence set EK of size K as the budget-bounded outcome of the same graph-formation process.** Rather than treating this as an independent ranking step performed over the activated documents, we formulate it as a *chain-integrity boundary* on Gq : EK should preserve the witnessed connectivity already established in Aq , and any document admitted into EK should either extend an existing witnessed chain or close a question-induced obligation that no document already in EK covers.

§4.3–§4.4 构造出的查询局部图 Gq 通常包含比 **reader budget K** 所能容纳的更多文档。在 **reader** 边界处，**EvidenceFlow** 把大小为 K 的证据集 **EK** 形成成为同一图形成过程的预算受限产物。我们不把它当作在被激活文档上独立进行的排序步骤，而是表述为 Gq 上的 **链完整性边界**： EK 应保留 **Aq** 中已经建立的被见证的连接，任何被准入 EK 的文档要么是延展一条已有的被见证的链，要么是闭合一个 **EK** 中现有文档尚未覆盖的、由问题诱导的义务。

§4.5.1 QUESTION-INDUCED OBLIGATIONS

For EvidenceFlow to determine whether a candidate document closes a part of q that the current EK does not yet cover, the chain-integrity boundary is conditioned on a small set of <i>question-induced obligations</i>: entity mentions and relation-bearing slots that q expects the evidence to jointly resolve. These obligations are extracted from q with the same controlled language model used for the question anchor in §4.3, applied to the question and not to the retrieved corpus, and do not generate evidence chains. Their role is purely local to the boundary decision: an obligation that is already addressed by an existing document in EK does not require a further admission, while an obligation still open after the current prefix becomes a precondition for admitting an additional document.	为了让 EvidenceFlow 能判断某个候选文档是否闭合了 q 中当前 EK 尚未覆盖的部分，链完整性边界条件作用于—小组 <i>问题诱导的义务</i>：即 q 期望证据共同解决的实体提及和承载关系的槽位。这些义务从 q 中抽取，使用与 §4.3 中问题锚点相同的受控语言模型——作用在问题上，不作用在检索到的语料上，也不生成证据链。它们的作用纯粹局部于边界决策：已经被 EK 中某个现有文档处理过的义务不需要进一步准入；而当前前缀之后仍然 open 的义务，则成为准入额外文档的先决条件。
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§4.5.2 WITNESS-PRESERVING FORMATION

Concretely, EK is formed incrementally from Gq in two interleaved modes. <i>Witness-preserving admission</i> extends EK along arcs of Aq, so that documents already supported by a witnessed transition to some $d \in EK$ are preferred to documents that would only enter EK through unwitnessed similarity. <i>Residual admission</i> is reserved for cases where an open obligation remains and a candidate document, not yet connected to EK by a witnessed transition, provides evidence for it. This second mode is what allows EK to cover the question's induced relation structure even when the witnessed component of Gq has not yet reached every required document. Residual admission does not create new unwitnessed arcs: once a document is admitted, any further movement involving it in Gq remains subject to the same witness conditions of §4.4.	具体地，EK 从 Gq 以两种交错模式增量形成。<i>Witness 保持的准入</i> 沿 Aq 中的弧延展 EK，使得已经通过被见证的 transition 连接到某个 $d \in EK$ 的文档，优于那些只能通过未被见证的相似度进入 EK 的文档。<i>残差准入</i> 仅保留用于以下情况：仍有未闭合的义务，且某个目前与 EK 之间还没有被见证的 transition 的候选文档能为该义务提供证据。这第二种模式使 EK 能够覆盖问题诱导的关系结构，即便 Gq 中被见证的部分尚未触及每一个所需的文档。<i>残差准入不会创建新的未被见证的弧</i>：一旦某文档被准入，Gq 中涉及它的任何后续移动仍然受 §4.4 中相同的 witness 条件约束。
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§4.5.3 BOUNDARY AS CONTINUATION OF GRAPH FORMATION

The chain-integrity boundary is not a separate scoring layer applied to a materialized candidate set. The same graph Gq and the same witness layer $\mathcal{F}$ that drove activation in §4.3–§4.4 also drive the boundary; the boundary is additionally conditioned on the obligation set extracted from q. EK is thus the reader-budget boundary of the same graph-formation process that built Gq, rather than an independently selected set produced by an external scoring procedure.	链完整性边界不是在已物化候选集上的独立打分层。在 §4.3–§4.4 中驱动激活的同一张图 Gq 和同一个 witness 层 $\mathcal{F}$ 也驱动边界；边界额外受从 q 抽取的义务集所制约。EK 因此是构造 Gq 的同一个图形成过程的预算边界，而不是由某个外部打分流程产生的独立选择集。
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